

[TTPs (Tactics, Techniques, and Procedures) - Security Operations Guide]

[What are TTPs?]

TTPs describe how cyber attackers plan, execute, and achieve their attack objectives. Security teams and SOC analysts use TTPs to understand attacker behavior, detect sophisticated attacks, perform threat hunting, improve incident response, and build resilient SIEM detection rules.

"TTPs explain the behavior and methods attackers use during cyber attacks."

[Breakdown of TTPs]

Term	Core Concept	Operational Meaning
Tactics	The Attacker's Goal	What the attacker wants to achieve.
Techniques	The Method Used	How the attacker intends to achieve that goal.
Procedures	The Exact Tool / Step	The precise technical execution step or tool deployment.

[TTP Behavioral Hierarchy]

└─ [Tactic] → What/Objective (e.g., Credential Access)

└─ [Technique] → How/Method (e.g., Credential Dumping)

└─ [Procedure] → Specific Implementation (e.g., Running Mimikatz)

[1. Tactics]

Tactics describe the high-level objectives an attacker aims to achieve during an active stage of a security incident campaign.

[Examples of Core Tactics]

Tactic Name	Defensive Purpose / Analytical Target
Initial Access	Methods used to enter the organizational network.
Execution	Running unauthorized malicious codes or tools on systems.
Persistence	Maintaining long-term access inside the system across reboots.
Privilege Escalation	Gaining administrative or root rights over system boundaries.
Defense Evasion	Avoiding active detection mechanics or purging telemetry logs.
Credential Access	Stealing corporate usernames, hashes, and cleartext passwords.
Discovery	Gathering internal host configuration or directory system information.
Lateral Movement	Moving from a compromised asset across internal servers or domains.
Collection	Gathering sensitive data arrays into a centralized point before exfiltration.
Exfiltration	Stealing files or moving secure internal data out of network borders.
Impact	Damaging configurations, manipulating systems, or deploying ransomware.

[2. Techniques]

Techniques represent the explicit conceptual methods that threat actors deploy to satisfy a specific parent tactic requirement.

[Common Adversary Techniques Matrix]

Technique	Defensive Description & Behavioral Manifestation
Phishing	Using spoofed email vectors to trick users into revealing tokens or clicking links.
PowerShell Abuse	Leveraging built-in OS command frameworks to download scripts directly into memory.
Credential Dumping	Extracting identity database hashes or secrets from system application memories.
DLL Injection	Injecting code structures directly into trusted running system processes.
Brute Force	Automating mass password guessing passes to compromise individual active accounts.
Command Execution	Running system-level terminal commands natively on target environments.
Spear Phishing	Crafting hyper-targeted social engineering communication loops aimed at single targets.

[3. Procedures]

Procedures represent the bottom granular layer of the framework. They describe the precise sequence of steps, scripting layouts, configuration parameters, or executable binary tools leveraged by an attacker.

[Comprehensive Practical TTP Mappings]	
Scenario A: Phishing Attack Vector	└─ Tactic: Initial Access └─ Technique: Phishing Email └─ Procedure: Sending fake corporate Microsoft login page link via external domains.
Scenario B: System Command Scripting	└─ Tactic: Execution └─ Technique: PowerShell Abuse └─ Procedure: Running command: [powershell.exe -EncodedCommand <Base64>]
Scenario C: Ransomware Operational Flow	└─ Tactic: Impact └─ Technique: Data Encryption └─ Procedure: Spawning ransomware binary executable to systematically encrypt files.
Scenario D: Target Credential Harvesting	└─ Tactic: Credential Access └─ Technique: Keylogging └─ Procedure: Silently side-loading a raw keylogger driver binary package on endpoints.

[Strategic Value & SOC Application]

Modern Security Operations Centers prioritize TTP detection frameworks over static indicators. Attackers can quickly change external source IPs, domains, or malware compilation file hashes (IOCs). However, changing their core structural execution behaviors (TTPs) is significantly harder and more expensive for them.

[Static Detection vs. TTP Behavior Hunting]	
└─ Static Signature Focus	→ Blocks specific malware hash (Easy for attackers to mutate)
└─ Behavioral TTP Focus	→ Detects [PowerShell downloading payload via Internet] (Hard

[TTP Monitoring Domains inside the SOC]

Analysts monitor distinct telemetry channels for TTP indicators: suspicious command-line strings, hidden PowerShell parameter inputs, anomalous concurrent log-in locations, credential dumping signatures, lateral file movements across domains, and system process modifications.

[TTP Detection Rules Mapping]

Suspicious Activity Signature	Correlated MITRE ATT&CK Tactic Category
Obfuscated / Base64 Encoded PowerShell Strings	Execution / Defense Evasion
Multiple rapid sequential failed login telemetry	Credential Access (Brute Force)
Unusual registry path writing or modifications	Persistence Mechanisms
Standard user account running administrative commands	Privilege Escalation Paths
Sustained, abnormal massive outbound network traffic	Exfiltration Channels

[The MITRE ATT&CK Knowledge Base]

The MITRE Corporation ATT&CK Framework is a globally accessible public repository organizing real-world threat actor TTPs. It empowers SOC teams, threat hunters, incident response crews, and security researchers to visualize coverage maps and understand adversary pathways cleanly.

[MITRE ATT&CK Grid Snippet]

Tactic Category	Correlated Technical Technique Examples
Execution	PowerShell Execution, Windows Command Shell, WMI Loops
Persistence	Registry Run Keys, Scheduled Tasks, Account Creation
Credential Access	Credential Dumping (LSASS Memory parsing), OS Credential Cracking

[Common Tools Used for TTP Hunting & Detection]

- **SIEM Ecosystems (Splunk):** Analyzing central log telemetry timelines for operational pattern changes.
- **EDR Suites (CrowdStrike / Microsoft Defender):** Monitoring real-time host behaviors and tracking process lineage anomalies.
- **Network Monitoring (Wireshark / Suricata):** Inspecting packet streams and network IDS rules for command signatures.
- **Windows Logging Frameworks (Sysmon):** Capturing granular host lifecycle events, process creations, and registry updates.

[Interview Preparation: Core Questions & Answers]

Q1. What do TTPs stand for and represent in Cybersecurity?

Answer: TTPs stand for Tactics, Techniques, and Procedures. They define the complete hierarchy of goals, methods, and precise steps deployed by threat actors to execute cyber campaigns.

Q2. What is the fundamental difference between a Tactic and a Technique?

Answer: A Tactic represents the overarching strategic objective or goal of an attacker during a phase. A Technique details the explicit behavioral method or path used to achieve that goal.

Q3. Give a clear, complete horizontal operational example of a TTP framework mapping.

Answer: Tactic: Credential Access. **Technique:** Credential Dumping. **Procedure:** Executing the Mimikatz binary tool on a Windows endpoint to scrape LSASS memory space.

Q4. Why is TTP-based behavior hunting more effective for a SOC than relying purely on IOC signatures?

Answer: Attackers can trivially alter file hashes and server IPs to bypass firewall blocks. However, altering internal behaviors—like writing code to memory or abusing administrative shells—requires rewriting their entire kit, making TTP-based hunting highly effective.

Q5. Which public documentation framework maps attacker TTPs across the lifecycle?

Answer: The globally recognized standard matrix framework is the **MITRE ATT&CK Matrix**.